Analyzing Discreet Differences: Understanding the Unique Traits of Magnetic and Electric Dipoles

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January 29, 2024

Abstract

Exploring classical electromagnetism, we discover surprising similarities between electric and magnetic fields, even in seemingly unrelated scenarios. Investigating these fields in interactive environments reveals similarities akin to non-interactive settings. However, a closer look reveals subtle yet significant differences. Our study delves into these distinctions between electric and magnetic dipoles, demonstrating their contrasting behaviors. Specifically, we highlight how electric dipoles can imitate monopole behavior, unlike their magnetic counterparts. Throughout our analysis, we consistently adhere to the "Steady Fields Assumption".

1 Conceptual Aspect of an Arbitrary Magnetization field

In discussing an arbitrary magnetization field $\vec{M}(\vec{r})$ in space, the concept involves placing infinitesimal magnetic dipoles, each with a moment $\delta\vec{m} = \vec{M}(\vec{r})\delta V$, across all spatial points. At first glance, it appears unnecessary to impose constraints on this magnetization field, labeling some instances as "non-physical" or "physically impossible." To validate this viewpoint, we can assemble the magnetization using the earlier discussed method of placing these dipole volume elements. If arbitrary moment vectors are deemed "physical" or "physically possible," then constructing any desired distribution of dipoles, leading to any arbitrary magnetization field, should be feasible. Consequently, any well-behaved magnetization field can be physically realized without any specific conditions, apart from maintaining its expected behavior. Moreover, the absence of singularities within the magnetic matter volume, though not a necessity, fortifies our theory by mitigating potential mathematical intricacies.

2 A Brief Review on The Usual Approach to Electric and Magnetic Field in Matter

To delve into Magnetization and examine the essential nuances between electric and magnetic dipoles, we are conducting a review of our investigations into these fields within matter.

2.1 Electric Field in Matter

When subjected to an increasing electric field, a non-conductive macroscopic body experiences an initial repositioning of its electrons within their fundamental structures. This rearrangement significantly impacts the overall multipole expansion of the body. The multipole expansion for a finite electric charge distribution, assuming stationarity, can be derived by applying Coulomb's law and Legendre polynomials for $1/|\vec{r} - \vec{r'}|$:

$$\phi = \sum_{l=0}^{\infty} \int_V \frac{\rho(\vec{r'})}{4\pi\epsilon_0 r} \left(\frac{r'}{r}\right)^l P_l(\cos\theta) d\tau' \tag{2.1}$$

For $|\vec{r}| \gg |\vec{r'}|$ the first term is called "monopole" which is

$$\phi_0 = \frac{1}{4\pi\epsilon_0 r} \int_V \rho(\vec{r'}) d\tau' \tag{2.2}$$

However, since electrons remain within the body and do not leave it, the total charge vanishes. Consequently, this term equates to zero. In cases where the body initially contains a charge, standard Laplace equation-solving methods can be employed for these charges. The second term in this expansion is referred to as the "dipole," which signifies...

$$\phi_1 = \frac{1}{4\pi\epsilon_0 r^2} \int_V \rho(\vec{r'}) r' \cos\theta d\tau' \tag{2.3}$$

Which can be written as

$$\phi_1 = \frac{\vec{p} \cdot \vec{r}}{4\pi\epsilon_0 r^3} \ , \ \vec{p} \equiv \int_V \rho(\vec{r'}) \vec{r'} d\tau' \tag{2.4}$$

Which the defined parameter $\vec{p}$ is called "dipole moment".

However, the subsequent terms ($l > 1$) can be neglected only under the condition that we are far enough from the body ($\vec{r} \gg \vec{r'}$), but this approximation fails for points in close proximity to the body. To accurately determine the electric field near the body, these neglected terms beyond the dipole contribution must be retained. Nevertheless, this approach remains an approximation and proves insufficient for determining fields within the body. Given our interest in solving the field throughout the entire space, a more comprehensive approach is necessary.

We've presented the multipole expansion for the entire body. An alternative approach involves formulating the multipole expansion "locally"; specifically for local structures such as atoms or molecules. In this context, the approximation ($\vec{r} \gg \vec{r'}$) holds well even within the body due to the typical size of local structures, which usually ranges from $\sim 10^{-10} - 10^{-6}\, m$. Given that electrons within non-conducting bodies reside within these local structures, the monopole term remains zero. Hence, our initial approximation for these materials involves local dipoles. Let's denote the dipole moment in a volume element dV as $d\vec{p} = \vec{P}dV$. The dipole density or "polarization" $\vec{P}$ can be defined as:

$$\vec{P}(\vec{r}) \equiv \frac{d\vec{p}}{dV} \tag{2.5}$$

The definition of polarization enables us to decompose the body into numerous local dipoles. Integrating their potential allows us to derive the electric potential and consequently the electric field at any point in space. This process involves substituting $\vec{p}$ with $\vec{P}dV$ in (2.4):

$$\phi = \int_V \frac{\vec{P}(\vec{r'}) \cdot \vec{\mathcal{R}}}{4\pi\epsilon_0 \mathcal{R}^3} d\tau' \,, \quad \vec{\mathcal{R}} \equiv \vec{r} - \vec{r'} \tag{2.6}$$

Utilizing the mathematical expression

$$\nabla \frac{1}{\mathcal{R}} = \frac{\vec{\mathcal{R}}}{\mathcal{R}^3} \,, \tag{2.7}$$

we can reframe Eq. (2.6) as follows:

$$\phi = \int_V \frac{\vec{P}(\vec{r'})}{4\pi\epsilon_0} \cdot \nabla \frac{1}{\mathcal{R}} \tag{2.8}$$

By applying the subsequent mathematical operations, we derive the following expression:

$$\begin{aligned} \phi &= \frac{1}{4\pi\epsilon_0} \int_V \vec{P}(\vec{r'}) \cdot \nabla \frac{1}{\mathcal{R}} d\tau' \\ \phi &= \frac{1}{4\pi\epsilon_0} \int_V \left(\nabla \cdot (\frac{\vec{P}(\vec{r'})}{\mathcal{R}}) - \frac{\nabla \cdot \vec{P}(\vec{r'})}{\mathcal{R}} \right) d\tau' \\ \phi &= \oint_{\partial V} \frac{\vec{P}(\vec{r'})}{4\pi\epsilon_0 \mathcal{R}} \cdot d\vec{a}' - \int_V \frac{\nabla \cdot \vec{P}}{4\pi\epsilon_0 \mathcal{R}} d\tau' \end{aligned} \tag{2.9}$$

We immediately notice the similarity between this equation and Coulomb's law

$$\phi = \oint_{\partial V} \frac{\sigma(\vec{r'})}{4\pi\epsilon_0 \mathcal{R}} da' + \int_V \frac{\rho(\vec{r'})}{4\pi\epsilon_0 \mathcal{R}} d\tau' \tag{2.10}$$

The polarization behaves akin to surface charge density and volume charge density, referred to as "bound charge". they can be obtained by equivalence of (2.9) and (2.10) yielding:

$$\boxed{\rho_b = -\nabla \cdot \vec{P}(\vec{r}) \quad \& \quad \sigma_b = \vec{P}(\vec{r}) \cdot \hat{n}} \tag{2.11}$$

"As demonstrated, substituting the dielectric body with specific charges enables us to derive the potential and electric field using conventional methods for solving Poisson and Laplace equations."

2.2 Magnetic Field in Matter

In the case of the magnetic field, while the equations for the electric field involve $\nabla \cdot \vec{E} = \rho/\epsilon_0$ and $\nabla \times \vec{E} = 0$, the magnetic field is described by $\nabla \cdot \vec{B} = 0$ and $\nabla \times \vec{B} = \mu_0 \vec{J}$. Due to the divergence of the field being zero and focusing on its curl, we anticipate the existence of a vector potential $\vec{A}$ such that $\nabla \times \vec{A} = \vec{B}$ which can be derived using Biot-Savart law and Eq. (2.7)

$$\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi} \int_V \frac{\vec{J}(\vec{r'}) \times \vec{\mathcal{R}}}{\mathcal{R}^3} d\tau' \tag{2.12}$$

$$\vec{B}(\vec{r}) = -\frac{\mu_0}{4\pi} \int_V \vec{J}(\vec{r'}) \times \nabla \frac{1}{\mathcal{R}} \, d\tau'$$

$$\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi} \int_V \nabla \times \left(\frac{\vec{J}(\vec{r'})}{\mathcal{R}} \right) d\tau' = \nabla \times \left(\frac{\mu_0}{4\pi} \int_V \frac{\vec{J}(\vec{r'})}{\mathcal{R}} d\tau' \right) \tag{2.13}$$

In the last step we used the fact that ∇ is taken on $\vec{r}$ while $\vec{J}(\vec{r'})$ is depended on $\vec{r'}$. As we expected, we successfully represented $\vec{B}(\vec{r})$ as $\nabla \times \vec{A}$ defining $\vec{A}$ as follows:

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int_V \frac{\vec{J}(\vec{r'})}{\mathcal{R}} d\tau' + \frac{\mu_0}{4\pi} \oint_{\partial V} \frac{d\vec{S}' \times \vec{K}(\vec{r'})}{\mathcal{R}} \tag{2.14}$$

The second term is added similarly to the second term in (2.10) for Coulomb's law, where $\vec{K}$ represents the surface current density.

Using this "Vector Potential", we eliminate pseudo-vectors and cross products that arise when deriving the magnetic field through the Biot-Savart equation, much like how we removed vectors when defining the electric potential. Similar to what we derived for the electric field, our objective is to derive $\vec{A}$ for a magnetic dipole by forming a multipole expansion for $\vec{A}$:

$$\vec{A}(\vec{r}) = \sum_{l=0}^{\infty} \frac{\mu_0}{4\pi} \int_V \frac{\vec{J}(\vec{r'})}{r} \left(\frac{r'}{r} \right)^l P_l(\cos\theta) d\tau' \tag{2.15}$$

the monopole term is

$$\vec{A}_0(\vec{r}) = \frac{\mu_0}{4\pi r} \int_V \vec{J}(\vec{r'}) d\tau' \tag{2.16}$$

Using Einstain summation convection, we obtain:

$$\frac{4\pi}{\mu_0} \vec{A}_0(\vec{r}) \cdot \hat{e_k} = \int_V J_k(\vec{r'}) d\tau' = \int_V J_i(\vec{r'}) \delta_{ik} d\tau' = \int_V J_i(\vec{r'}) \partial'_i r'_k d\tau'$$

$$\int_V J_k(\vec{r'}) d\tau' = \int_V \left(\partial'_i (J_i r'_k) - r'_k \partial'_i J_i \right) d\tau' = \oint_{\partial V} r'_k J_i dS_i - \int_V r'_k \partial'_i J_i d\tau'$$

$$\int_V \vec{J}(\vec{r'}) d\tau' = \oint_{\partial V} \vec{r'} \left(\vec{J}(\vec{r'}) \cdot d\vec{S} \right) - \int_V \vec{r'} \left(\nabla' \cdot \vec{J}(\vec{r'}) \right) d\tau' \tag{2.17}$$

The first and second terms are eliminated due to localization and stationarity assumptions, respectively.

$$\vec{A}_0(\vec{r}) = 0 \tag{2.18}$$

As we've demonstrated, the presence of a monopole term is impossible with steady localized current arrangements. Given the absence of a monopole term, it appears feasible to employ the method used for dielectric polarization to observe a similar phenomenon in magnetism. Let's derive the dipole term for $\vec{A}$ from Eq. (2.14):

$$\begin{aligned}
\vec{A}_1(\vec{r}) &= \frac{\mu_0}{4\pi r^2}\int_V \vec{J}(\vec{r'})r'\cos\theta d\tau' \\
\vec{A}_1(\vec{r})\cdot\hat{e_k} &= \frac{\mu_0\vec{r}}{4\pi r^3}\cdot\int_V J_k\vec{r'}d\tau' = \frac{\mu_0}{4\pi r^3}\sum_j r_j\int_V J_k r'_j d\tau'
\end{aligned} \tag{2.19}$$

The assumption of localized current states that:

$$\begin{aligned}
\int_V \nabla\cdot\left(f(\vec{r})\vec{J}(\vec{r})\right)d\tau &= \oint_{\partial V} f(\vec{r})\vec{J}(\vec{r})\cdot d\vec{S} = 0 \\
\int_V \left(f(\vec{r})\nabla\cdot\vec{J}(\vec{r}) + \vec{J}(\vec{r})\cdot\nabla f(\vec{r})\right)d\tau &= 0
\end{aligned} \tag{2.20}$$

Substituting $f(\vec{r}) = r_i$ yields (2.18). When we use $f(\vec{r}) = r_j r_k$, conservation of charge for steady arrangements, expressed as $\nabla\cdot J = 0$, allows us to utilize (2.20):

$$\begin{aligned}
\sum_i\int_V J_i\partial_i(r_j r_k)d\tau &= \sum_i\int_V \left(J_i r_k\partial_i r_j + J_i r_j\partial_i r_k\right)d\tau = 0 \\
\sum_i\int_V \left(J_i r_k\delta_{ij} + J_i r_j\delta_{ik}\right)d\tau &= \int_V \left(J_j r_k + J_k r_j\right)d\tau = 0 \\
\int_V \left(J_j r_k - J_k r_j\right)d\tau &= \int_V \left(J_j r_k + J_k r_j - 2J_k r_j\right)d\tau = -2\int_V J_k r_j d\tau \\
\int_V J_k r_j d\tau &= -\frac{1}{2}\int_V \left(J_j r_k - J_k r_j\right)d\tau = -\frac{1}{2}\int_V \left(\vec{r}\times\vec{J}\right)_k \epsilon_{ijk}d\tau
\end{aligned} \tag{2.21}$$

The equation (2.21) holds for any arbitrary components i, j. Simplifying (2.19) using (2.21):

$$\begin{aligned}
\vec{A}_1(\vec{r}) &= \frac{\mu_0}{4\pi r^3}\sum_j -\frac{1}{2}r_j\int_V \left(\vec{r'}\times\vec{J}\right)_k \epsilon_{ijk}d\tau \\
\vec{A}_1(\vec{r}) &= \frac{\mu_0}{4\pi r^3}\left(-\frac{1}{2}\vec{r}\times\int_V \vec{r'}\times\vec{J}d\tau\right) \\
\vec{A}_1(\vec{r}) &= \frac{\mu_0}{4\pi}\frac{\vec{m}\times\vec{r}}{r^3}\ ,\ \vec{m} \equiv -\frac{1}{2}\int_V \vec{r'}\times\vec{J}(\vec{r'})d\tau
\end{aligned} \tag{2.22}$$

Where $\vec{m}$ is known as the "magnetic dipole moment", similar to $\vec{P}$ for electric potential. Taking the curl of both sides of (2.22) and comparing it to the $\nabla\phi_1$ derived in (2.4):

$$\nabla \times \vec{A}_1 = \vec{B}_1(\vec{r}) = \frac{\mu_0}{4\pi}\left(\frac{3\left(\vec{m}\cdot\hat{r}\right)\hat{r} - \vec{m}}{r^3}\right) \tag{2.23}$$
$$-\nabla\phi_1 = \vec{E}_1(\vec{r}) = \frac{1}{4\pi\epsilon_0}\left(\frac{3(\vec{p}\cdot\hat{r})\hat{r} - \vec{p}}{r^3}\right)$$

It's clear that $\vec{E}_1$ and $\vec{B}_1$ behave identically throughout space. Building on our understanding from dielectrics, let's explore magnetism within matter by defining "Magnetization" as the density of magnetic dipoles moment:

$$\vec{M}(\vec{r}) = \frac{d\vec{m}}{dV} \tag{2.24}$$

Using this definition, we derive equivalent of (2.10) for magnetization:

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi}\int_V \vec{M}(\vec{r'}) \times \nabla\left(\frac{1}{\mathcal{R}}\right) d\tau'$$
$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi}\int_V \left(\frac{\nabla\times\vec{M}(\vec{r'})}{\mathcal{R}} - \nabla\times\left(\frac{\vec{M}(\vec{r'})}{\mathcal{R}}\right)\right) d\tau'$$
$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi}\int_V \frac{\nabla\times\vec{M}(\vec{r'})}{\mathcal{R}} d\tau' - \frac{\mu_0}{4\pi}\oint_{\partial V}\frac{\vec{M}(\vec{r'})}{\mathcal{R}} dS' \tag{2.25}$$

And equivalent of (2.11) according to (2.14):

$$\boxed{\vec{J}_b(\vec{r}) = \nabla\times\vec{M}(\vec{r}) \quad \& \quad \vec{K}_b(\vec{r}) = \vec{M}(\vec{r})\times\hat{n}} \tag{2.26}$$

Using this approach, we're able to use our previous methods of poisson equation solving in order to solve electric and magnetic fields inside matter utilizing some effective currents and charges (bound charge and bound current) stemming from polarization and magnetization, respectively.

2.3 Displacement and H Vectors

Going further through steady electromagnetism theory inside matter, we derive maxwell equations in steady form for arbitrary media:

$$\nabla\cdot\vec{E} = \frac{\rho_{tot}}{\epsilon_0} = \frac{\rho_b + \rho_f}{\epsilon_0} = \frac{\rho_f}{\epsilon_0} - \frac{\nabla\cdot\vec{P}}{\epsilon_0}$$
$$\nabla\cdot\vec{D} = \rho_f\ ,\ \vec{D} \equiv \vec{P} + \epsilon_0\vec{E} \tag{2.27}$$

Here, we've defined free charge density ρ_f which is the real electric charge. in other words, $\vec{D}$ serves as an alternative to $\vec{E}$ but discounting the impact of bound charges.

There is a similar parameter for magnetism:

$$\nabla \times \vec{B} = \mu_0 \vec{J}_{tot} = \mu_0 \left(\vec{J}_f + \vec{J}_b \right) = \mu_0 \vec{J}_f + \mu_0 \nabla \times \vec{M}$$

$$\nabla \times \vec{H} = \vec{J}_f \ , \ \vec{H} \equiv \frac{\vec{B}}{\mu_0} - \vec{M} \tag{2.28}$$

By employing these definitions, we achieve:

$$\nabla \cdot \vec{H} = -\nabla \cdot \vec{M} \tag{2.29}$$

$$\nabla \times \vec{D} = \nabla \times \vec{P} \tag{2.30}$$

3 A Contradiction

In this instance, we encounter a contradiction implying $\nabla \cdot \vec{B} \neq 0$, conflicting with established electromagnetism principles and Maxwell's equations. Our aim is to scrutinize this contradiction, emphasizing nuances to demonstrate how diligent attention to detail resolves any apparent inconsistencies. Our primary focus revolves around understanding how a diverging polarization induces monopole behavior, whereas diverging magnetization fails to produce a similar effect.

3.1 The Idea

Results (2.29) and (2.30) approve the validity of $\nabla \cdot \vec{B} = 0$ and $\nabla \times \vec{E} = 0$ within steady fields, regardless of polarization and magnetization.However, despite their similarities, our methodologies for studying $\vec{B}$ and $\vec{E}$ inside matter took distinct paths. This disparity originates from our focus on investigating field potentials rather than the vector fields themselves. What if our derivation of the magnetic field involved integrating over a magnetic dipole field while seeking a polarization-like theory for magnetization?

We perform integration over $\vec{B}_1$ according to (2.23), substituting $\vec{m}$ with $\vec{M}(\vec{r'})d\tau'$:

$$\vec{B}(\vec{r}) = \int_V \frac{\mu_0}{4\pi} \left(\frac{3\left(\vec{M}(\vec{r'}) \cdot \hat{\mathcal{R}}\right)\hat{\mathcal{R}} - \vec{M}(\vec{r'})}{\mathcal{R}^3} \right) d\tau'$$

$$\vec{B}(\vec{r}) = -\nabla \int_V \frac{\mu_0}{4\pi} \frac{\vec{M}(\vec{r'}) \cdot \vec{\mathcal{R}}}{\mathcal{R}^3} d\tau' = -\nabla \int_V \frac{\mu_0 \vec{M}(\vec{r'})}{4\pi} \cdot \nabla' \left(\frac{1}{\mathcal{R}} \right) d\tau'$$

$$\vec{B}(\vec{r}) = -\nabla \int_V \left(\nabla' \cdot \left(\frac{\mu_0 \vec{M}(\vec{r'})}{4\pi\mathcal{R}} \right) - \frac{\mu_0 \nabla' \cdot \vec{M}(\vec{r'})}{4\pi\mathcal{R}} \right) d\tau'$$

$$\vec{B}(\vec{r}) = -\nabla \left(\oint_{\partial V} \frac{\mu_0 \vec{M}(\vec{r'})}{4\pi\mathcal{R}} \cdot d\vec{S}' - \int_V \frac{\mu_0 \nabla' \cdot \vec{M}(\vec{r'})}{4\pi\mathcal{R}} d\tau' \right) \tag{3.1}$$

In this context, the operation of ∇ pertains to the components of $\vec{r}$ rather than those of $\vec{r'}$. Consequently, when taking the divergence of $\vec{B}(\vec{r})$:

$$\nabla \cdot \vec{B}(\vec{r}) = \int_V \frac{\mu_0 \nabla' \cdot \vec{M}(\vec{r'})}{4\pi} \nabla^2 \left(\frac{1}{\mathcal{R}}\right) d\tau' - \oint_{\partial V} \frac{\mu_0}{4\pi} \nabla^2 \left(\frac{1}{\mathcal{R}}\right) \vec{M}(\vec{r'}) \cdot d\vec{S'}$$

$$\nabla \cdot \vec{B}(\vec{r}) = \oint_{\partial V} \mu_0 \delta(\vec{\mathcal{R}}) \vec{M}(\vec{r'}) \cdot d\vec{S'} - \int_V \mu_0 \nabla' \cdot \vec{M}(\vec{r'}) \delta(\vec{\mathcal{R}}) d\tau' \tag{3.2}$$

In the last step, we have used $\nabla^2(1/\mathcal{R}) = -4\pi\delta^3(\vec{\mathcal{R}})$. Our focus lies on evaluating $\nabla \cdot \vec{B}$ within the material. To accomplish this, we position $\vec{r}$ inside the substance, causing the first term to vanish due to $\delta^3(\vec{\mathcal{R}}) = 0$ for all points along the surface ($\vec{\mathcal{R}}$ always possesses a positive length for $\vec{r}$ within and $\vec{r'}$ on the boundary of the material). Simplifying the recent outcome:

$$\boxed{\nabla \cdot \vec{B}(\vec{r}) = -\mu_0 \nabla \cdot \vec{M}(\vec{r})} \tag{3.3}$$

Which is a complete contradiction with our discussion on arbitrary magnetization field in section 1. The discrepancy emerges when we apply Maxwell's equations and establish a condition regarding $\vec{M}$:

$$\nabla \cdot \vec{B} = 0 \Rightarrow \nabla \cdot \vec{M} = 0 \tag{3.4}$$

3.2 First Attempt

To address the contradiction that surfaced, our focus shifts to the magnetic field on a single dipole itself, which, naturally, tends towards infinity. However, we introduce a corrective term, $\vec{\alpha}(\vec{m})\delta(\vec{r})$, carefully adjusted to nullify the problematic term in (3.3) and consequently resolve the contradiction. Now, let's reconsider the expression for the magnetic field of a single dipole:

$$\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi} \left(\frac{3\left(\vec{m} \cdot \hat{r}\right)\hat{r} - \vec{m}}{r^3} + \vec{\alpha}(\vec{m})\delta^3(\vec{r}) \right) \tag{3.5}$$

Through the operations performed to derive (3.2) for this modified field, we obtain:

$$\nabla \cdot \vec{B}(\vec{r}) = \oint_{\partial V} \mu_0 \delta^3(\vec{\mathcal{R}}) \vec{M}(\vec{r'}) \cdot d\vec{S'} - \int_V \mu_0 \nabla' \cdot \vec{M}(\vec{r'}) \delta^3(\vec{\mathcal{R}}) d\tau' + \frac{\mu_0}{4\pi} \nabla \cdot \int_V \vec{\alpha}(\vec{r'}) \delta^3(\vec{\mathcal{R}}) d\tau'$$

$$\nabla \cdot \vec{B}(\vec{r}) = \oint_{\partial V} \mu_0 \delta^3(\vec{\mathcal{R}}) \vec{M}(\vec{r'}) \cdot d\vec{S'} - \int_V \mu_0 \nabla' \cdot \vec{M}(\vec{r'}) \delta^3(\vec{\mathcal{R}}) d\tau' + \frac{\mu_0}{4\pi} \nabla \cdot \vec{\alpha}(\vec{r})$$

$$\nabla \cdot \vec{B}(\vec{r}) = -\mu_0 \nabla \cdot \vec{M}(\vec{r}) + \frac{\mu_0}{4\pi} \nabla \cdot \vec{\alpha}(\vec{r}) \tag{3.6}$$

Now, by selecting $\vec{\alpha}(\vec{r}) = 4\pi \vec{M}(\vec{r})$, we resolve the contradiction and maintain $\nabla \cdot \vec{B} = 0$. Thus, the modified magnetic field of a single dipole is given by:

$$\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi} \left(\frac{3\left(\vec{m} \cdot \hat{r}\right)\hat{r} - \vec{m}}{r^3} + 4\pi \vec{m} \delta^3(\vec{r}) \right) \tag{3.7}$$

By incorporating this adjusted field, we have ensured $\nabla \cdot \vec{B} = 0$. Implementing this modification up to (3.1) and utilizing the definition of $\vec{H}$ in (2.28), we obtain:

$$\begin{aligned} \vec{B}(\vec{r}) &= -\nabla \left(\oint_{\partial V} \frac{\mu_0 \vec{M}(\vec{r'})}{4\pi \mathcal{R}} \cdot d\vec{S}' - \int_V \frac{\mu_0 \nabla' \cdot \vec{M}(\vec{r'})}{4\pi \mathcal{R}} d\tau' \right) + \mu_0 \vec{M}(\vec{r}) \\ \vec{H}(\vec{r}) &= -\nabla \left(\oint_{\partial V} \frac{\vec{M}(\vec{r'})}{4\pi \mathcal{R}} \cdot d\vec{S}' - \int_V \frac{\nabla' \cdot \vec{M}(\vec{r'})}{4\pi \mathcal{R}} d\tau' \right) \end{aligned} \tag{3.8}$$

Taking the divergence of both sides, akin to (3.3), we arrive at a similar result for $\vec{H}$, which is precisely the equation (2.29):

$$\nabla \cdot \vec{H} = -\nabla \cdot \vec{M} \tag{2.29}$$

3.3 Electric Field Case

One might inquire about the outcome if we apply a similar approach to an electric dipole.

$$\begin{aligned} \vec{E}(\vec{r}) &= \int_V \frac{1}{4\pi\epsilon_0} \left(\frac{3\left(\vec{P}(\vec{r'}) \cdot \hat{\mathcal{R}}\right)\hat{\mathcal{R}} - \vec{P}(\vec{r'})}{\mathcal{R}^3} \right) d\tau' \\ &= \nabla \times \int_V \frac{1}{4\pi\epsilon_0} \frac{\vec{P}(\vec{r'}) \times \vec{\mathcal{R}}}{\mathcal{R}^3} d\tau' = \nabla \times \int_V \frac{1}{4\pi\epsilon_0} \vec{P}(\vec{r'}) \times \nabla' \left(\frac{1}{\mathcal{R}} \right) d\tau' \\ &= \frac{1}{4\pi\epsilon} \nabla \times \left[\int_V \nabla' \times \left(\frac{\vec{P}(\vec{r'})}{\mathcal{R}} \right) d\tau' - \int_V \frac{\nabla' \times \vec{P}(\vec{r'})}{\mathcal{R}} d\tau' \right] \\ &= \frac{1}{4\pi\epsilon_0} \nabla \times \left[\oint_{\partial V} \frac{\vec{P}(\vec{r'}) \times d\vec{S}'}{\mathcal{R}} - \int_V \frac{\nabla' \times \vec{P}(\vec{r'})}{\mathcal{R}} d\tau' \right] \\ &= \frac{1}{4\pi\epsilon_0} \nabla \times \int_V \frac{\nabla' \times \vec{P}(\vec{r'})}{\mathcal{R}} d\tau' = \frac{1}{4\pi\epsilon_0} \int_V \left(\nabla' \times \vec{P}(\vec{r'}) \right) \times \nabla \frac{1}{\mathcal{R}} d\tau' \\ \nabla \times \vec{E}(\vec{r}) &= \frac{1}{4\pi\epsilon_0} \int_V \left[\left(\nabla' \times \vec{P}(\vec{r'}) \right) \left(\nabla^2 \frac{1}{\mathcal{R}} \right) - \left(\nabla' \times \vec{P}(\vec{r'}) \cdot \nabla \right) \nabla \frac{1}{\mathcal{R}} \right] \\ &= -\frac{1}{4\pi\epsilon_0} \int_V \left(\nabla' \times \vec{P}(\vec{r'}) \right) 4\pi \delta^3(\vec{\mathcal{R}}) d\tau' \end{aligned}$$

$$\boxed{\nabla \times \vec{E}(\vec{r}) = -\frac{\nabla \times \vec{P}(\vec{r})}{\epsilon_0}} \tag{3.9}$$

Which also leads to a contradiction similar to the magnetism case. Our attempt to address this issue results in:

$$\vec{E}(\vec{r}) = \int_V \frac{1}{4\pi\epsilon_0} \left(\frac{3\left(\vec{P}(\vec{r'}) \cdot \hat{\mathcal{R}}\right)\hat{\mathcal{R}} - \vec{P}(\vec{r'})}{\mathcal{R}^3} + \vec{\beta}(\vec{r'})\delta^3(\vec{\mathcal{R}}) \right) d\tau'$$

$$\nabla \times \vec{E}(\vec{r}) = -\frac{\nabla \times \vec{P}(\vec{r})}{\epsilon_0} + \frac{1}{4\pi\epsilon_0} \nabla \times \vec{\beta}(\vec{r})$$

$$\vec{\beta}(\vec{r}) = 4\pi\vec{P} \tag{3.10}$$

This term is similar to what we derived for the magnetic dipole.

3.4 Understanding Incompleteness: The Path to Comprehensive Solution

It appears that the vector field

$$\vec{v}(\vec{r}) = \begin{cases} \frac{3\left(\vec{X}\cdot\hat{r}\right)\hat{r} - \vec{X}}{4\pi r^3}, & r > 0 \\ \vec{X}\delta^3(\vec{r}) & r = 0 \end{cases} \tag{3.11}$$

where $\vec{X}$ is a constant vector, is both divergence-free and curl-free throughout the entire space. Let's examine the Dirac delta term by integrating $\vec{v}(\vec{r})$ over the entire space. Utilizing identity (2.17) for $\vec{v}$ instead of $\vec{J}$:

$$\int_V \vec{v}(\vec{r}) d\tau = \int_{\partial V} \vec{r}\left(\vec{v} \cdot d\vec{S}\right) - \int_V \vec{r}\, \nabla \cdot \vec{v}\, d\tau$$

As we extend V to infinity, we have the flexibility to shape it as needed. Here, we consider V as a sphere with a radius of R centered at the origin. Examining the first term in spherical coordinates, the Dirac delta vanishes as $\vec{r}$ is always non-zero:

$$\vec{v}(\vec{r}) = \frac{|\vec{X}|}{4\pi r^3}\left(2\cos\theta\ \hat{r} + \sin\theta\ \hat{\theta}\right)$$

$$\int_{\partial V} \vec{r}\left(\vec{v} \cdot d\vec{S}\right) = \int_{\partial V} \vec{r}\ \frac{|\vec{X}|}{2\pi r^3} \cos\theta\ r^2 \sin\theta\ d\theta\ d\phi$$

Utilizing cylindrical symmetry around the $\vec{X}$ axis, we obtain:

$$\begin{aligned}
\int_{\partial V} \vec{r}\left(\vec{v}\cdot d\vec{S}\right) &= \int_{\partial V} r\cos\theta\,\frac{\vec{X}}{2\pi r^3}\cos\theta\; r^2\sin\theta\; d\theta\; d\phi \\
&= \int_{\partial V} \frac{\vec{X}}{2\pi}\cos^2\theta\sin\theta\; d\theta\; d\phi \\
&= \vec{X}\int_0^{\pi}\cos^2\theta\sin\theta\; d\theta \\
\int_{\partial V} \vec{r}\left(\vec{v}\cdot d\vec{S}\right) &= \frac{2}{3}\vec{X} \qquad (3.12)
\end{aligned}$$

This holds true for both magnetic and electric dipoles ($\vec{X} = \vec{p}/\epsilon_0$ for electric and $\vec{X} = \mu_0\vec{m}$ for magnetic dipole). However, upon closer examination of the second term, we observe a distinction:

$$\int_V \vec{r}\,\nabla\cdot\vec{E}\; d\tau = \frac{1}{\epsilon_0}\int_V \vec{r}\,\rho(\vec{r})\; d\tau = \frac{\vec{P}}{\epsilon_0}$$

$$\int_V \vec{r}\,\nabla\cdot\vec{B}\; d\tau = 0$$

We've derived

$$\int_{space} \vec{E}d\tau = -\frac{1}{3}\frac{\vec{p}}{\epsilon_0} \qquad (3.13)$$

$$\int_{space} \vec{B}d\tau = \frac{2}{3}\mu_0\vec{m} \qquad (3.14)$$

Let's derive them from fields we've made for electric and magnetic dipoles.

$$\begin{aligned}
\int_V \vec{v}(\vec{r})d\tau &= \int_V \left[\frac{3\left(\vec{X}\cdot\hat{r}\right)\hat{r} - \vec{X}}{4\pi r^3} + \vec{X}\delta^3(\vec{r})\right] d\tau \\
&= \int_V \frac{|\vec{X}|}{4\pi}\left(\frac{2\cos\theta\;\hat{r} + \sin\theta\;\hat{\theta}}{r^3}\right) d\tau + \vec{X} \\
&= \vec{X} + \int_{r\to 0^+}^{R}\int_{\theta=0}^{\pi}\frac{|\vec{X}|}{2r^3}\left(3\cos^2\theta - 1\right) r^2\sin\theta\; d\theta\; dr \\
&= \vec{X}
\end{aligned}$$

This equation implies that

$$\int_{space} \vec{E}d\tau = \frac{\vec{p}}{\epsilon_0} \qquad (3.15)$$

$$\int_{space} \vec{B}d\tau = \mu_0\vec{m} \qquad (3.16)$$

This presents a clear contradiction with (3.13) and (3.14). So, what is the underlying problem, and how can we obtain a flawless solution?

3.5 A Definitive and Precise Resolution: Unveiling Mathematical Details in Carefully Addressing Paradoxes

The use of potentials, ϕ for the electric field and $\vec{A}$ for the magnetic field, becomes advantageous when taking derivatives to derive the respective fields. While working with the electric potential ϕ, $\nabla \times \vec{E} = 0$ is automatically satisfied without the need for additional precautions in our calculations. Similarly, when dealing with $\vec{A}$, the condition $\nabla \cdot \vec{B} = 0$ is inherently maintained. This is why we encountered no issues while developing the electromagnetism theory inside matter using potentials.

However, when directly dealing with fields, it becomes crucial to be mindful of fields Dirac delta functions in our calculations. To address this, we've devised a method. Consider the case of taking the derivative of a function, $f(x)$, where Dirac delta terms are significant. We first identify the part of f that may lead to a Dirac delta term. Then, by introducing a variable and avoiding this possibility, we examine the limit of the result as this variable approaches zero. To illustrate this method, let's apply it to finding the gradient of the electric dipole scalar potential:

$$\phi = \frac{\vec{X} \cdot \vec{r}}{4\pi r^3}$$

$$-\nabla\phi = \lim_{\Delta \to 0} \left[\nabla \frac{\vec{X} \cdot \vec{r}}{4\pi (r^2 + \Delta)^{\frac{3}{2}}} \right] = \frac{1}{4\pi} \lim_{\Delta \to 0} \left[\frac{3(\vec{X} \cdot \vec{r})\vec{r} - r^2 \vec{X}}{(r^2 + \Delta)^{\frac{5}{2}}} - \frac{\vec{X}\Delta}{(r^2 + \Delta)^{\frac{5}{2}}} \right]$$

$$-\nabla\phi = \frac{1}{4\pi} \lim_{\Delta \to 0} \left[\frac{3(\vec{X} \cdot \vec{r})\vec{r} - r^2 \vec{X}}{(r^2 + \Delta)^{\frac{5}{2}}} \right] - \frac{1}{4\pi} \lim_{\Delta \to 0} \left[\frac{\vec{X}\Delta}{(r^2 + \Delta)^{\frac{5}{2}}} \right]$$

Where the second term can be expressed as a Dirac delta function (this can be verified through integration and the axioms of Dirac delta) as:

$$\delta^3(\vec{r}) = \lim_{\varepsilon \to 0} \left[\frac{3\varepsilon}{4\pi (r^2 + \varepsilon)^{\frac{5}{2}}} \right]$$

Utilizing this identity, we obtain the **Curl-Free Dipole Field**:

$$-\nabla \left(\frac{\vec{X} \cdot \hat{r}}{4\pi r^2} \right) = \frac{3(\vec{X} \cdot \hat{r})\hat{r} - \vec{X}}{r^3} - \frac{1}{3}\vec{X}\delta^3(\vec{r}) \tag{3.17}$$

Applying the same method to the vector potential of a magnetic dipole, we obtain the **Divergence-Free Dipole Field**:

$$\nabla \times \left(\frac{\vec{X} \times \hat{r}}{4\pi r^2} \right) = \frac{3(\vec{X} \cdot \hat{r})\hat{r} - \vec{X}}{r^3} + \frac{2}{3}\vec{X}\delta^3(\vec{r}) \tag{3.18}$$

Finally, we've obtained a comprehensive solution for the electric and magnetic

dipole fields:

$$\boxed{\vec{E}_1(\vec{r}) = \frac{3(\vec{p}\cdot\hat{r})\hat{r} - \vec{p}}{r^3} - \frac{1}{3}\frac{\vec{p}}{\epsilon_0}\delta^3(\vec{r})} \tag{3.19}$$

$$\boxed{\vec{B}_1(\vec{r}) = \frac{3(\vec{m}\cdot\hat{r})\hat{r} - \vec{m}}{r^3} + \frac{2}{3}\mu_0\vec{m}\delta^3(\vec{r})} \tag{3.20}$$

4 A Problem with a Familiar Answer

- Consider a spherical medium uniformly polarized by a constant polarization vector $\vec{P}$. Determine the electric field at the center of this polarized sphere.
- We determine the bound charges and integrate over them:

$$\sigma_b = \vec{P}\cdot\hat{n} = |\vec{P}|\cos\theta$$
$$\vec{E} = -\hat{P}\int \frac{P\cos\theta R\cos\theta}{4\pi\epsilon_0 R^3} R^2 \sin\theta\, d\theta\, d\phi = -\hat{P}\int_0^\pi \frac{P}{2\epsilon_0}\cos^2\theta\sin\theta\, d\theta$$
$$\vec{E} = -\frac{1}{3}\frac{\vec{P}}{\epsilon_0} \tag{4.1}$$

There is also a similar problem in magnetism as well.

- Consider a spherical medium uniformly magnetized by a constant magnetization vector $\vec{M}$. Determine the magnetic field at the center of this magnetized sphere.
- We can utilize the last solution for magnetism but there is another simpler method with the same result. We take advantage of similarities:

$$\vec{H} = -\frac{1}{3}\vec{M}$$
$$\vec{B} = \mu_0\left(\vec{M} + \vec{H}\right)$$
$$\vec{B} = \frac{2}{3}\mu_0\vec{M} \tag{4.2}$$

Answers (4.1) and (4.2) are independent of the radius of the sphere. Therefore, they remain unchanged under the limit when R tends to zero. The concept of the field at the center when tending R to zero is the field of a point dipole on itself, which is exactly the result we obtained in (3.19) and (3.20). This remarkable compatibility showcases the strength of the mathematical methods we've employed to construct our theory.

5 The Final Instance

One might consider a problem involving divergent magnetization and draw parallels with polarization, thinking it might lead to a monopole term at that point. Let's delve into this problem.

- Consider a spherical magnetic matter with radius R including the magnetization
$$\vec{M}(r) = \alpha \frac{\hat{r}}{r^2}$$
where α, β are constant. Investigate the magnetic field in space.

- We start with equations for $\vec{H}$ and then derive $\vec{B}$:
$$\nabla \cdot \vec{H} = -\nabla \cdot \vec{M} = -4\pi\alpha\delta^3(r)$$
$$\nabla \times \vec{H} = 0\,,\ symmetry \rightarrow \vec{H} = H(r)\hat{r}$$
$$\int_V \nabla \cdot \vec{H} d\tau = \oint_{\partial V} \vec{H} \cdot d\vec{S} \rightarrow -\int_V 4\pi\alpha\delta^3(r) d\tau = 4\pi r^2 H(r)$$
$$\vec{H}(r) = -\frac{\alpha}{r^2}\hat{r}\int_V \delta^3(r) d\tau = -\alpha\frac{\hat{r}}{r^2}$$
Which is held through whole space. To find $\vec{B}$ inside the sphere we obtain:
$$\vec{B}(r) = \mu_0\left(\vec{H} + \vec{M}\right)$$
So the magnetic field throughout space is
$$\vec{B}(r) = 0 \quad , \quad r < R$$
$$\vec{B}(r) = -\mu_0\alpha\frac{\hat{r}}{r^2} \quad , \quad r > R$$
Where doesn't cause any contradiction with whatever we've used to build-up our theory.

6 Conclusion

Electric and magnetic dipoles, while sharing many similarities, exhibit distinct characteristics reflected in an additional term within their respective fields, represented by Dirac delta functions. Although this term has no impact on the outside fields, its significance becomes apparent within matter, where a dipole is always present at the point under consideration. The electric and magnetic dipole fields are:
$$\vec{E}_1(\vec{r}) = \frac{3(\vec{p}\cdot\hat{r})\hat{r} - \vec{p}}{r^3} - \frac{1}{3}\frac{\vec{p}}{\epsilon_0}\delta^3(\vec{r})$$
$$\vec{B}_1(\vec{r}) = \frac{3(\vec{m}\cdot\hat{r})\hat{r} - \vec{m}}{r^3} + \frac{2}{3}\mu_0\vec{m}\delta^3(\vec{r})$$
The disparity in their Dirac delta terms serves as the fundamental origin for any distinctions observed in their field behavior.

References

- Classical Electrodynamics, J. D. Jackson, 1999
- Introduction to Electrodynamics, D. J. Griffiths, 1999